# Hints: Abstract algebra – 4. Group actions, homomorphisms and automorphisms

Homework

*The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.*

---

## Problem 1

**Hint 1.** Two easy facts: $N_{UVW} = N_{VWU} = N_{WUV}$, since $xyz = e \iff yzx = e$; and $N_{UVG} = |U|\,|V|$, since $z$ is determined by $x, y$.

**Hint 2.** Expand $|A|\,|B| = N_{ABG} = N_{ABA} + N_{ABB} + N_{ABC}$ and $|B|\,|A| = N_{BAG} = N_{BAA} + N_{BAB} + N_{BAC}$. The first two terms match by cyclic invariance, so $N_{ABC} = N_{BAC}$, which is $N_{CBA}$ by cyclic invariance.

## Problem 2

**Hint 1.** Let $T = \{gh, gh^{-1}, g^{-1}h, g^{-1}h^{-1}\}$. You must show that every element is a product of between 1 and $|G|$ elements of $T$. Let $S$ be the set of all finite products of elements of $T$. In a finite group, a set closed under multiplication is a subgroup.

**Hint 2.** $S$ contains $(gh)(g^{-1}h)^{-1} = g^2$. Since $\operatorname{ord}(g)$ is odd, $g \in \langle g^2\rangle \subseteq S$, then $h = g^{-1}(gh) \in S$. So $S = G$.

**Hint 3.** For the bound, let $U_r$ be the set of products of at most $r$ elements of $T$, so $U_{r+1} = T \cup U_r T$. If $U_{r+1} = U_r$, the chain is stationary from then on. Before that it grows strictly, starting from $|U_1| \geqslant 1$, so $U_{|G|} = S = G$.

## Problem 3

*Note.* As printed, the bound $p^{n/p}$ is false: $G = (\mathbb{Z}/2)^2$ has $n = 4$, $p = 2$ and $16 > 4$ endomorphisms. The intended bound (Romania 2008) is $\sqrt[p]{n^n} = n^{n/p}$; the hints prove that.

**Hint 1.** An endomorphism is determined by the images of a generating set. Choose generators greedily, $g_{i+1} \notin \langle g_1, \ldots, g_i\rangle$. Each step raises the order of the generated subgroup by a factor that is at least a prime, so $G$ has a generating set with $k \leqslant \Omega(n)$ elements, where $\Omega$ counts prime factors with multiplicity.

**Hint 2.** So $|\operatorname{End} G| \leqslant n^k \leqslant n^{\Omega(n)}$. Write $n = pm$; then $\Omega(n) = 1 + \Omega(m) \leqslant 1 + (m-1) = n/p$, because $\Omega(m) \leqslant \log_2 m \leqslant m - 1$.

**Hint 3.** Equality needs $\Omega(m) = m - 1$, i.e. $m \in \{1, 2\}$, and every choice of images of the generators must extend. So $n = p$ or $n = 2p$ (with $p$ the largest prime). Check the candidates: $\mathbb{Z}/p$ has $p = n^1$ endomorphisms; for $n = 4$, $(\mathbb{Z}/2)^2$ has $16 = 4^2$ and $\mathbb{Z}/4$ only 4; for $n = 2p$ with $p$ odd, $\mathbb{Z}/2p$ has $2p$ and $D_p$ has $p^2 + 1$, both less than $4p^2$. Equality: $\mathbb{Z}/p$ and $(\mathbb{Z}/2)^2$.

## Problem 4

**Hint 1.** Let $d = \gcd(m, n) = sm + tn$, $A = G(m)$, $B = G(n)$. Both are normal (generated by conjugation-invariant sets). Since $x^d = (x^m)^s (x^n)^t$, you get $G(d) = \langle A, B\rangle$.

**Hint 2.** For $a \in A$, $b \in B$ the commutator $[a, b]$ lies in $A \cap B$, by normality of each. Elements of $A \cap B$ commute with $A$ and with $B$, hence are central in $H = G(d)$. So $H$ has nilpotency class at most 2, and the commutator is bimultiplicative on $H$.

**Hint 3.** Let $p = x^d$, $q = y^d$, and $m = dm'$, $n = dn'$. Then $p^{m'}, q^{m'} \in A$ commute, so $[p,q]^{m'^2} = e$; likewise $[p,q]^{n'^2} = e$. Since $\gcd(m'^2, n'^2) = 1$, $[p,q] = e$. So the generators of $H$ commute.

## Problem 5

**Hint 1.** An endomorphism of $\mathbb{Z}/n$ is $x \mapsto ax$ with $a = f(1)$, and it is bijective iff $\gcd(a,n) = 1$. An additive map of $(\mathbb{Z}/p)^k$ is $\mathbb{F}_p$-linear. $|\mathrm{GL}_k(\mathbb{F}_p)| = \prod_{i=0}^{k-1}(p^k - p^i)$: choose the columns one by one outside the span of the previous ones.

**Hint 2.** If $|\operatorname{Aut} G| = 1$, then $\operatorname{Inn} G \cong G/Z(G)$ is trivial, so $G$ is abelian. Then $x \mapsto x^{-1}$ is an automorphism, so $x^2 = e$ for all $x$.

**Hint 3.** So $G \cong (\mathbb{Z}/2)^k$ with $\mathrm{GL}_k(\mathbb{F}_2)$ trivial, i.e. $k \leqslant 1$: $G$ is trivial or $\mathbb{Z}/2$.

## Problem 6

**Hint 1.** The rotation group of the cube has 24 elements: the identity; 6 rotations by $\pm 90°$ about face axes; 3 by $180°$ about face axes; 8 by $\pm 120°$ about vertex axes; 6 by $180°$ about edge axes. For each type, count the cycles on faces and on vertices.

**Hint 2.** Faces: $\frac{1}{24}(k^6 + 3k^4 + 12k^3 + 8k^2)$. Vertices: $\frac{1}{24}(k^8 + 17k^4 + 6k^2)$. Check: for $k = 2$ these give 10 and 23.

**Hint 3.** Integrality follows from Burnside: the expression counts orbits. A polynomial with non-integral coefficients can still be integer-valued, like $\binom{k}{2}$.

## Problem 7

**Hint 1.** Let $e$ be the identity of $(G, *)$. For any $a$, $a * e = a$, while $a \times e \perp a$. So either $a = 0$ or $a \times e = 0$. If $e \neq 0$, every element is parallel to $e$ and all cross products vanish. So assume $e = 0$.

**Hint 2.** Suppose $a \times b \neq 0$, so $c = a * b = a \times b$. Compute $(a * a) * b = a * c = a \times c$ and $(a * a) * c = a * (a \times c) = a \times (a \times c) = -|a|^2 c$. Neither is orthogonal to $b$, resp. $c$.

**Hint 3.** So $(a * a) \times b = 0$ and $(a * a) \times c = 0$, i.e. $a * a$ is parallel to both $b$ and $c$, hence $a * a = 0 = e$. Then $b = (a * a) * b = a \times c = a \times (a \times b)$, which forces $a \perp b$ and $b = -|a|^2 b$: impossible.

## Problem 8

**Hint 1.** Let $F(g)$ be the number of fixed points and $n = |X|$. Burnside gives $\sum_g F(g) = |G|$. Applying it to the action on $X \times X$ (at least two orbits: the diagonal and its complement) gives $\sum_g F(g)^2 \geqslant 2|G|$.

**Hint 2.** Consider $\sum_g (F(g) - 1)(F(g) - n)$. It equals $\sum F^2 - |G| \geqslant |G|$. Each term with $1 \leqslant F(g) \leqslant n$ is $\leqslant 0$, and each $g$ without fixed points contributes exactly $n$.

**Hint 3.** Hence $n \cdot \#\{g : F(g) = 0\} \geqslant |G|$, so the proportion is at least $1/n$, in particular positive.

## Problem 9

**Hint 1.** $\varphi \circ i_g \circ \varphi^{-1} = i_{\varphi(g)}$.

**Hint 2.** Two transpositions commute iff they are disjoint (otherwise their product has order 3). So $\varphi(1\,2), \ldots, \varphi(1\,n)$ are pairwise non-commuting transpositions. For $n \geqslant 4$, show they all share one common point $a$ (three pairwise intersecting transpositions either share a point or form a triangle $(xy), (yz), (xz)$, and a fourth one cannot meet all three sides of a triangle).

**Hint 3.** Write $\varphi(1\,i) = (a\,b_i)$ and define $\sigma(1) = a$, $\sigma(i) = b_i$. Then $\varphi$ and conjugation by $\sigma$ agree on the generators $(1\,i)$. ($n \leqslant 3$: check directly.)

## Problem 10

**Hint 1.** The action on the $n$ cosets gives $G \to S_n$. Its kernel $K$ is normal, and $G/K$ embeds in $S_n$, so $[G : K]$ divides $n!$.

**Hint 2.** If a simple group of order 60 had a subgroup of index $k$ with $1 < k < 5$, the kernel would be a proper normal subgroup, hence trivial. Then $G$ would embed in $S_k$, and $60 \mid k!$ is impossible for $k \leqslant 4$.